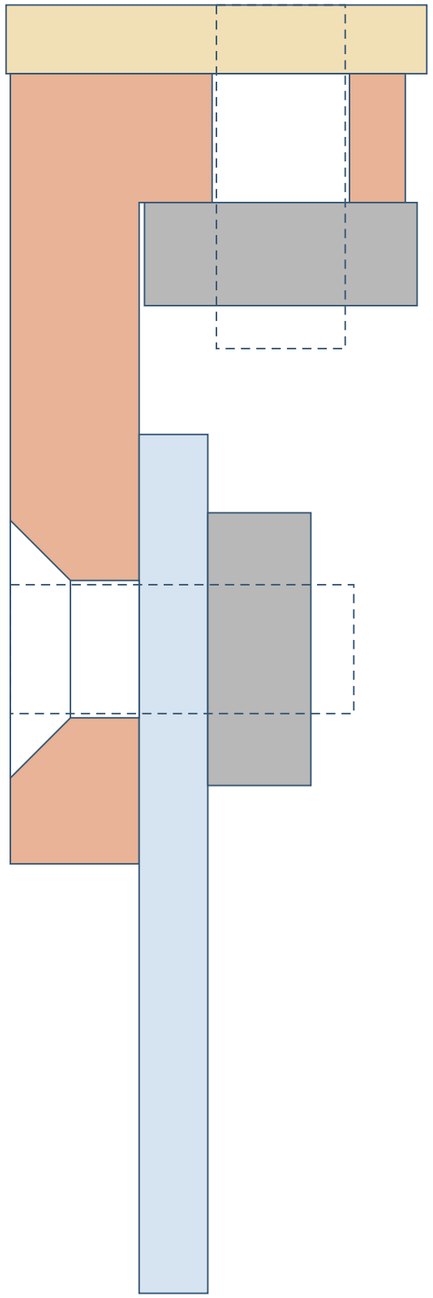
A4.1 | PLAIN L BRACKET - PRINT TWO IDENTICAL PARTS

3mm walls, two holes, no nut traps or inserts. ABS prototype. M3x8 front + M3x8 countersunk side; four loose M3 nuts.

XZ SECTION THROUGH BOTH HOLES



Local origin = assembly X0.1 / Z1.6
Overall 9.2 X x 10 Y x 18.4 Z
BOTH legs are 3.0mm thick.
Upright on PCB BACK face (B.Cu X3.1).
Plain nuts sit OUTSIDE the printed L.
No slots, captive pockets, roofs or inserts.

HOLES / LOCAL DIMENSIONS
Front: X6.3 / Y5.0, diameter3.2
Side: Y5.0 / Z13.4, diameter3.2
Side countersink: diameter6.0 x 90 deg
Depth1.4; leaves1.6 of straight bore.
Countersink PLASTIC ONLY, never FR4.

FASTENERS
Front M3x8 pan/button, head <=6 dia
Side M3x8 flat countersunk (90 deg)
Side length includes head; seat it flush.
4 loose plain M3 nuts: AF<=5.5, h<=2.4
No washers or tall nyloc nuts.
Minimum nominal thread exit: 1.0 each.

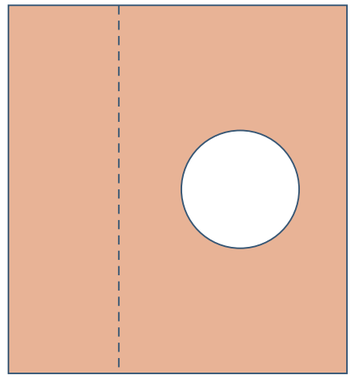
FIT LIMITS
Front nut-to-upright: only 0.125 nominal
Nut swept right edge X9.575 (panel9.8)
Side head flush to0.3mm below face
Side screw tip X8.1..8.4
Measure ABS shrinkage and actual nuts.

SERVICE OUT OF RACK: hold front nut with small pliers from +X; side nut with 5.5mm nut driver from +X.
Drive screws from panel front (-Z) and bracket outer side (-X). Soldered PCB can stay fitted.
Print L profile flat. ABS is a candidate, not strength approval. Trial screw seating, nut/tool access and shrinkage.

HOLE COORDINATES / MOUNTING

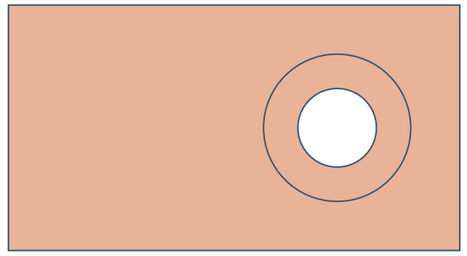
Panel: X6.4, Y22 / Y106.5, diameter3.2
Main PCB: same Y, Z15, diameter3.2
Bracket pitch:84.5; panel holes moved +1.5 X.
PCB plane and main mounting holes unchanged.

FRONT FACE / LOCAL XY



Hole (6.3,5.0)
9.2 x 10 face
Dashed: 3mm wall

OUTER SIDE / LOCAL YZ



Hole (Y5.0,Z13.4); outer circle is countersink.

Source: hardware/mechanics/corner_braces.scad

SCAD only; no tested STL or physical print.